



Composite Functions - Practice (and solutions)

For the given functions f and g , find (answer on the back)

(a) $(f \circ g)(x)$

(b) $(g \circ f)(x)$

(c) $(f \circ f)(x)$

(d) $(g \circ g)(x)$

1. $f(x) = 2x + 3, \quad g(x) = 3x$

2. $f(x) = \sqrt{x}, \quad g(x) = x^2$

3. $f(x) = \frac{x+1}{x-1}, \quad g(x) = \frac{x-1}{x+1}$

4. $f(x) = x + \frac{1}{x}, \quad g(x) = x^2$

For each of the following problems, show that $(f \circ g)(x) = (g \circ f)(x) = x$.

1. $f(x) = 2x, \quad g(x) = \frac{1}{2}x$

2. $f(x) = ax + b, \quad g(x) = \frac{1}{a}(x - b), \quad a \neq 0$

3. $f(x) = \frac{1}{x}, \quad g(x) = \frac{1}{x}$

4. $f(x) = \frac{2x+1}{x-1}, \quad g(x) = \frac{x+1}{x-2}$

Answers

1 a) $f(g(x)) = 2(3x) + 3 = 6x + 3$ 2 a) $f(g(x)) = \sqrt{(x^2)} = x$

b) $g(f(x)) = 3(2x + 3) = 6x + 9$ b) $g(f(x)) = (\sqrt{x})^2 = x$

c) $f(f(x)) = 2(2x + 3) + 3 = 4x + 9$ c) $f(f(x)) = \sqrt{\sqrt{x}} = \sqrt[4]{x}$